

Structure of an Atom

Chemistry plays an important role in NEET after Biology. Comparatively, Chemistry is the most scoring subject if prepared thoroughly. Therefore, to master Chemistry, the

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points and principles. Memorise the formulas and important equations because a little confusion in any of these can ruin your entire answer. While going through the chapter thoroughly, it is advisable to make flow charts, tables, concept maps and diagrams to make it easier for you to revise in the future.

Flow of Historical Events Contributing to the Discovery of Structure of Electrons

Structure of an atom is basic yet the most intimidating concept which involves the historical facts and derivations of the discovery of atoms along with the recent studies. It is one of the most lengthy chapters and includes Grassroot topics that form the basis of Chemistry as a whole. In the early 4th and 5th century Democritus and Leucippus came up with the idea that everything in the world is made up of "atoms" which are uncuttable. In 1803, Dalton stated in his theories that everything on earth is a matter which is made up of tiny, indestructible and indivisible unit particles called atoms. Along with other assumptions, he also stated that atoms cannot be broken down further as it is the smallest particle.

In 1885, Sir William Crookes discovered electrons by cathode ray experiment and thus Dalton's statement that atoms are indivisible was proved wrong. William confirmed through his experiments that the particles in the cathode rays are negatives and it is present in all the atoms.

In 1904 J. J Thomson studied William's experiment and came up with the Plum Pudding Model to explain the structure of atoms. According to his model, an atom is electrically neutral as it is a positive ball of a sphere with negative electrons embedded side. He had no idea about the presence of protons then, but in 1886 his model failed because Eugen Goldstein discovered protons in similar way electrons were discovered. He concluded that anode rays consist of positive particles called protons and it is also present in all the atoms. Therefore, the discovery of electrons and protons were done by 1886 but the exact structure of the atom was not known.

In 1911, Ernest Rutherford performed an alpha scattering experiment where a stream of alpha particles was bombarded on the gold foil and thus the nucleus was discovered. So, according to Rutherford's model, an atom consists of a nucleus at the center which contains positively charged protons and neutral neutrons. Electrons, the negatively charged particles revolve around the nucleus. There were three reasons for the failure of Rutherford's model and one of them was pointed out by Maxwell in his electromagnetic theory where he explained that if a charged particle moves around another charged particle then it loses energy. So, if electrons revolve around the nucleus then it can follow a spiral path and fall into the nucleus. In 1913, Niels Bohr corrected Rutherford's model by explaining that electrons move in a definite circular path of fixed energy around the centre of the nucleus.

Study of Electrons and Nucleons

Atoms can be divided into two regions, nuclear and extranuclear. The nuclear region consists of a nucleus which has positively charged particles called protons and neutral particles called neutrons. The mass of a proton and a neutron are almost the same, that is, 1.67×10^{-27} . Electrons are the negatively charged particles revolving around the nucleus in a fixed orbit. Since the mass of the electron is almost negligible, the entire weight of the atom is because of protons and neutrons present inside the nucleus. Atomic Number (Z) of an atom is the number of protons present in the nucleus. As we know, an atom as a whole is neutral that means the number of electrons and protons are the same. Therefore we can say that atomic number is the number of protons as well as the number of electrons present in an atom. Mass Number (A) is the number of nucleons of the atom that is the sum of the number of protons and neutrons. The other subatomic particles found are positron, neutrino, antineutrino, meson, antiproton and ν -particles.

Electromagnetic Radiations

In 1870, James Maxwell stated that an electrically charged particle, when moved under acceleration, produces alternating electrical and magnetic fields that are mutually perpendicular to each other. This electrical and magnetic field are transmitted in the form of wavelengths, frequency, speed and amplitude. These waves are called electromagnetic waves.

Planck's Quantum Theory

In 1901, Max Planck concluded that radiant energy is emitted or absorbed only in discrete units in the form of bundles or packets of energy called photons or quantum. Photon is just a bundle of energy and has no mass. In vacuum, the speed of electromagnetic waves is the same as the speed of light.

There are few terms associated with waves. These are wavelength, frequency, velocity, amplitude and wave number. Wavelength is the length of one complete wave, that is the distance between successive crests and troughs. Frequency is the number of waves passing through a point per second. Wave velocity is the distance travelled by wave per second. Amplitude is the maximum displacement of the highest point in the trough from its mean position.

Photoelectric Effect

The photoelectric effect is the phenomenon of ejection of electrons from the surface of metal if a light of a specific range of frequency strikes the surface of the metal. The minimum frequency of the light that causes the ejection of electrons is called Threshold frequency. To release electrons, the metal surface absorbs a part of photon energy. This is known as the work function of the metal surface.

Quantum Mechanical Model

The Quantum Mechanic Model is based on the two principles – Broglie's Principle and Heisenberg's principle. In 1920, Schrodinger studied the behaviour of electrons and described them in terms of the Schrodinger wave equation. The equations are single-valued, finite and continuous and the solutions of this equation are called Eigenvalues.

Aufbau Principle and Pauli's Exclusion Principle

Electrons fill up the orbital according to their increasing order of energies that is 1s, 2s, 2p, 3s, 4s, 3d, 4p, 5s, 4d and it goes on. According to the Aufbau Principle, the order of the energy can be calculated by $(n+l)$ rule, and thus the energy of the atomic orbitals depends on the value of n . Hund's rule states that the pairing of the opposite spins of electrons does not take place unless and until all the subshells are singly occupied such that orbitals having single electrons have the same spins. Pauli's exclusion principle states that in an atom only two electrons having opposite spin can occupy the same orbital.

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